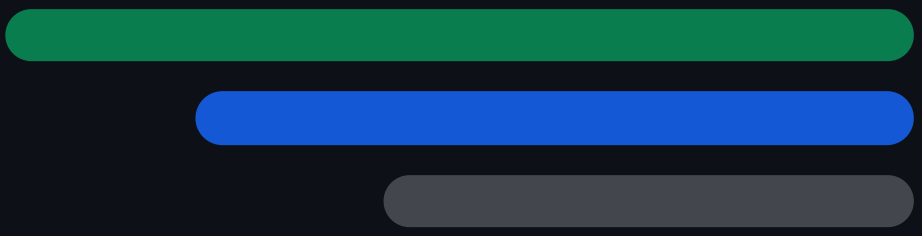
ULTIMATE APP · BORACAY

# Technical Documentation

Architecture, modules, money flows, integrations, deployment and testing of the Ultimate App platform.



| | |
|---|---|
| Document | Technical Documentation, edition 2026.10 |
| Date | 13 October 2026 |
| Platform | PHP 8.1+ · MySQL / MariaDB (InnoDB) · Progressive Web App |
| Hosting | Hostinger (shared hosting, HTTPS required) |
| Source | ailevoler/serialnumber · `ultimateapp-agents-portal/` |
| Audience | Developers, system administrators, finance and operations |

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# 1 System overview

Ultimate App is a mobile-first super-app for Boracay. One PHP code base and one MySQL database serve five web applications: the customer app, the URide Driver App, the Merchant Portal, the Agents Portal and the Admin Panel.

| 227 | ~20,100 | 58 | 23 |
|---|---|---|---|
| PHP files (excluding tests and vendor) | lines of PHP | database tables | automated test suites |

## 1.1 Applications and users

| Application | URL | Users | Main purpose |
|---|---|---|---|
| **Customer app** (PWA) | `/` | Residents and tourists | Wallets (Credits, BCash), URide, UPass/UGo/ULocal/UEat, bills and load, QR payments, exchange, UBarangay, KYC, Face Login |
| **URide Driver App** (PWA) | `/driver/` | Drivers, boat operators, vehicle owners | Sign-up, go online, receive and run rides and rentals, self-drive hand-over, wallet, top-ups, payouts |
| **Merchant Portal** (PWA) | `/merchant-portal/` | Businesses | Onboarding, payment QR, listings and orders, UEat restaurant, payouts, MCTC top-up center |
| **Agents Portal** | `/agent-portal/` | Referral agents | Referral link/QR, Master and Sub-Agent teams, commissions, payouts |
| **Admin Panel** | `/admin/` | Super Admin, Finance, Support, Barangay Officer | CRM, ERP (payments, finance, ledger, settlements), URide, marketplace, KYC, settings, audit log |

## 1.2 Technology stack

| Area | Technology |
|---|---|
| Server language | PHP 8.1 or newer (tested on 8.3), `declare(strict_types=1)` in libraries; no framework |
| Database | MySQL / MariaDB, InnoDB, `utf8mb4`; PDO with exceptions and prepared statements |
| Front end | Server-rendered HTML, plain CSS and vanilla JavaScript; installable PWA (service worker, manifests) |
| Maps | Google Maps JavaScript API, Places API (New), Geocoding API (optional) |
| Face and liveness | MediaPipe Face Landmarker (Apache-2.0) and face-api (MIT), both self-hosted under `assets/vendor/` |
| QR codes | qrcode-generator 1.4.4 (MIT), html5-qrcode 2.3.8 (Apache-2.0) |
| Payments in | PayMongo QR Ph (Payment Intents and Hosted Checkout, signed webhooks) |
| Bills, load, e-wallet cash-in | SUNIWAY Partner API |

| Area | Technology |
|---|---|
| Exchange rates | Frankfurter (ECB) with currency-api fallback, or a manual rate |
| Time zone | Asia/Manila in PHP and `+08:00` in every MySQL session |

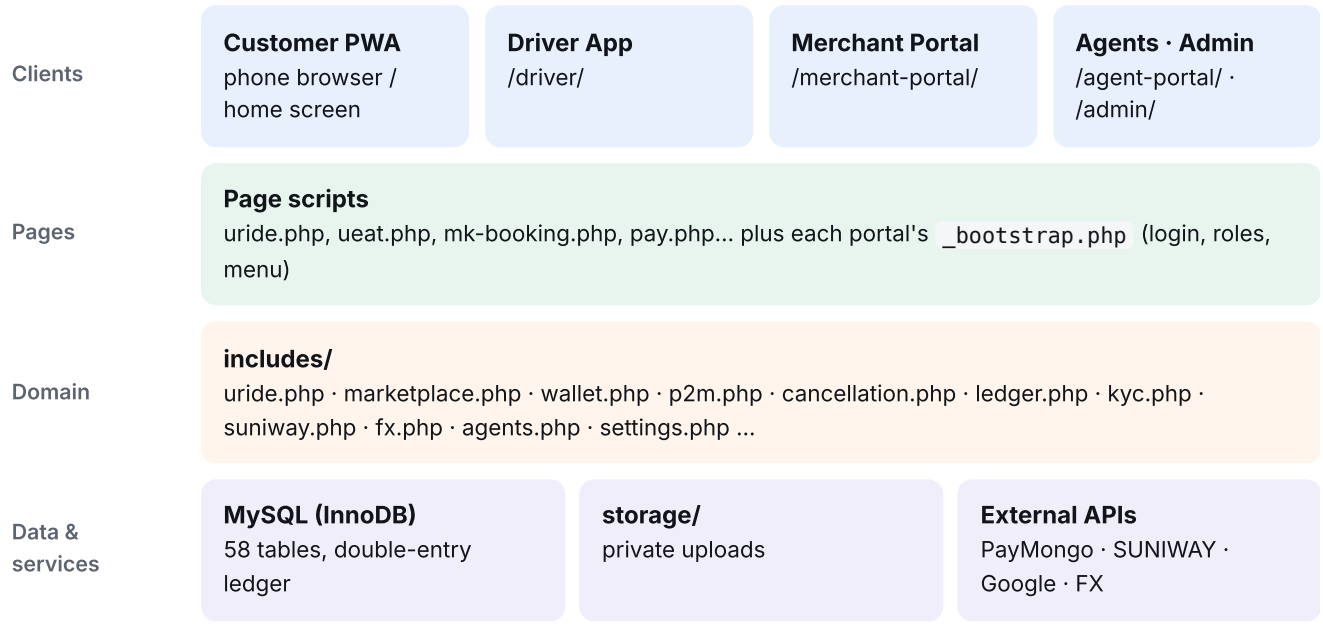
## 1.3 Customer services

| Service | What it does | Paid with |
|---|---|---|
| **URide** | Point-to-point rides (Motorcycle, E-Trike, Car, Van, **Boat**), Rent a Ride with driver, Self-drive | Credits, BCash, Cash* |
| **UPass** | Airport transfers, island/day passes, event passes, entrance fees | Credits, BCash, on the day* |
| **UGo** | Tours and activities with time slots | Credits, BCash, on the day* |
| **ULocal** | Local services (spa, photography, laundry, guides…) | Credits, BCash, on the day* |
| **UEat** | Food delivery and pick-up from merchant restaurants | Credits, BCash |
| **UBills / ULoad / UCash In** | Bills payment, prepaid load, send to GCash/Maya and other e-wallets | Credits |
| **Pay / Scan QR** | Pay merchants (P2M) and people (P2P) | Credits, BCash |
| **Exchange** | Buy and sell USD and about 20 other currencies | BCash |
| **UBarangay** | Barangay permits and certificates with QR verification (switchable) | Credits, BCash, QR Ph |

* Cash and pay-on-the-day are available only while Admin keeps **Pay Cash** on (chapter 4).

# 2 Architecture

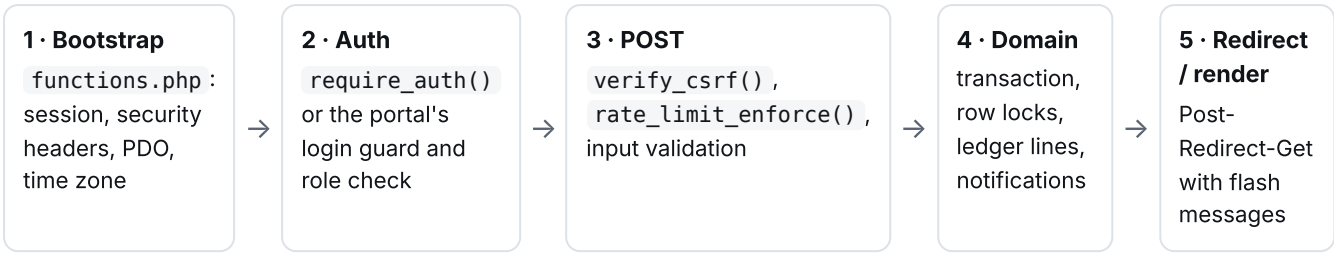
A classic page-per-file PHP application. Each page includes a shared bootstrap, does its own POST handling (CSRF-checked), calls domain functions in `includes/`, and renders HTML. All business rules live in the domain libraries so that pages, cron jobs and tests use the same code.




## 2.1 Folder layout

| Path | Contents |
|---|---|
| `/*.php` | Customer pages (dashboard, uride, ueat, upass/ugo/ulocal, pay, bcash-*, kyc, face-login, ubills...) |
| `includes/` | Domain libraries and shared layout ( `functions.php` bootstrap, `header.php`, `footer.php`, `menu.php` ) |
| `admin/` | Admin Panel; `_bootstrap.php` defines roles, permissions and menu |
| `driver/` | URide Driver App (own login, manifest, JSON `api.php` for live feed and GPS) |
| `merchant-portal/` | Merchant Portal (onboarding, listings, orders, payouts, MCTC top-up) |
| `agent-portal/` | Agents Portal (referrals, team, earnings, payouts) |
| `assets/` | CSS, JavaScript, images, PWA icons, `vendor/mediapipe` (~22 MB), `vendor/face-api` (~14 MB) |
| `config/` | `config.php` (DB), `paymongo.php`, `google_maps.php`, `app_secret.php` (generated encryption key), `storage.php` (optional) |
| `database/` | `schema.sql` and 37 migration files, plus `ALL_UPDATES_2026-09.sql` |
| `storage/` | Private uploads: `kyc/`, `licenses/`, `driver_docs/`, `market/`, `brgy/`, `cache/` (blocked from the web) |
| `cron/` | `fx-refresh.php` (hourly exchange-rate refresh) |
| `tests/` | CLI test suites and fake provider servers (never deployed) |

## 2.2 Request flow




Domain functions throw `InvalidArgumentException` for user-facing errors (shown as alerts) and other exceptions for system faults (logged, generic message shown). Every money change runs inside one InnoDB transaction.

## 2.3 Settings

- Stored in `app_settings` (`setting_key`, `setting_value`, `is_secret`). Read with `setting()` / `setting_int()`; written with `setting_save()`, which records the admin and audit trail.
- `includes/settings.php` holds the schema: type, limits and label for every key (e.g. `uride.boat.per_km_centavos`, `cancel.ueat_pct`).
- Secret values (PayMongo and SUNIWAY keys) are encrypted with the key in `config/app_secret.php`. Back up that file with the site, or the secrets must be entered again.
- Most settings are edited in **Admin › Settings** (Super Admin only): Payments, Charges & limits, Google Maps, URide.

## 2.4 File storage

Uploads are saved on disk, not in the database, with random names, and served only through access-checked PHP scripts (e.g. `market-photo.php`, `admin/kyc-file.php`). Images are re-encoded on upload, which strips EXIF/GPS data. To keep files safe across redeploys, copy `config/storage.php.example` to `config/storage.php` and point it to a folder outside `public_html`. **Admin › System › Files & storage** runs a write test and lists records whose files are missing.

# 3 Money model and ledger

Two customer wallets, one rule: every money movement is posted to a balanced double-entry ledger in the same database transaction as the balance change.

**Credits**

The main in-app balance. **1 Credit = ₱1.00.** Stored in `users.credits` (decimal pesos). Bought with QR Ph or at an MCTC top-up center; spent on every service. Ledger account `user_credits:ID`.

**BCash**

A separate peso wallet in `boracay_cash_wallets.balance_centavos`. Converted 1:1 from/to Credits with no fee; used for transfers, payments and currency exchange. Ledger account `user_cash:ID`.

## 3.1 Amounts

- All new money columns are **integer centavos** (`*_centavos`). No floating-point arithmetic is used for money.
- Percentages are computed with integer basis points and rounded once (e.g. commission `intdiv(fare × bp + 5000, 10000)`).
- URide fares and rental prices are rounded **up** to the whole peso.

## 3.2 The ledger

`ledger_post($pdo, $group, $event, $lines)` inserts lines into `ledger_entries`. Each line is `[account, debit, credit]`. The function refuses any entry whose debits and credits differ, or that is zero. The `group` ties the lines to a business reference (e.g. a ride code, `RF-…` for a refund, `CF-…` for a cancellation fee).

| Account | Meaning |
|---|---|
| `user_credits:ID` | Customer Credits |
| `user_cash:ID` | Customer BCash |
| `merchant_payable:ID` | Owed to a merchant (paid out in Settlements) |
| `driver_wallet:ID` | URide driver wallet |
| `uride_escrow` | URide fares on hold until the trip ends |
| `uride_deposit` | Self-drive deposits held |
| `platform_sales:<service>` | Ultimate App sales (its own UPass/UGo/ULocal listings) |
| `fee_revenue:<kind>` | Charge and commission revenue |
| `agent_payable:ID · agent_commission_expense` | Agent commissions owed and their expense |
| `paymongo_clearing · bank_clearing` | Money in transit with PayMongo or the bank |
| `settlement_payable:…` | Payouts requested but not yet transferred |
| `mctc_merchant_due:ID · mctc_collections:ID · office_cash` | Cash collected for top-ups |
| `suniway_payable` | Owed to SUNIWAY for bills, load and cash-in |

| Account | Meaning |
|---|---|
| `fx_usd_book · fx_book:<ccy> · fx_fee_revenue` | Customer foreign-currency holdings (PHP value) and exchange fees |
| `brgy_payable:<barangay>` | Permit fees owed to each barangay |
| `adjustments` | Manual admin adjustments |

**Admin › Finance** shows liabilities, revenue, a trial balance and reconciliation checks (for example, every driver wallet and every merchant MCTC due must equal its ledger balance). **Admin › General ledger** lists and exports entries.

## 3.3 Safety against double charging

- **Request keys.** Booking and payment forms carry a random one-time key; replays return the original result instead of charging again.
- **Row locks.** Balances, rides, orders and top-ups are read with `SELECT … FOR UPDATE` inside the transaction (e.g. first driver to accept wins).
- **Unique constraints.** Payment IDs, references, receipt numbers per merchant and approval keys are unique in the database.
- **Webhooks** are verified by signature and matched to the stored order (amount, currency, mode, session) before settling.

# 4 Payment methods

Admin controls which payment methods customers see. Credits are always on; BCash and Pay Cash have their own switches.

**Admin › Settings › Charges & limits › Payment methods**

| Method | Switch | Covers |
|---|---|---|
| **Credits** | Always on | Every service |
| **BCash** | `pay_bcash_enabled()` | URide (rides, rentals, self-drive), UPass, UGo, ULocal, UEat, QR payments |
| **Pay Cash** | `pay_cash_enabled()` | Cash to the URide driver; "pay on the day" for UPass, UGo and ULocal |

- When Pay Cash is off, every payment is held in the app and refunded automatically on cancel, decline or expiry.
- URide has its own Credits/Cash/BCash switches; a method shows only when both the app-wide and the URide switch are on.
- Existing bookings keep the method they were made with.

## 4.1 Implementation

| Function (includes/wallet.php) | Purpose |
|---|---|
| `wallet_debit_tx($pdo,$uid,$source,$amount,$title,$evenWhenOff=false)` | Debit Credits or BCash inside the caller's transaction; refuses insufficient balance. `$evenWhenOff` lets a cancellation fee be taken even if BCash payments were switched off later. |
| `wallet_credit_tx(…)` | Credit (refund) the same wallet |
| `wallet_account() · wallet_balance() · wallet_label()` | Ledger account name, balance and display label for a source |
| `p2m_pay_tx() · p2m_refund_tx() (includes/p2m.php)` | Pay a merchant from Credits or BCash with the configured charge, and refund it in full |

## 4.2 Merchant charges

Paying a merchant (QR or marketplace) applies the charge in **Charges & limits**: a percentage plus a fixed amount, separately for Credits and BCash, and who pays it (customer = added on top; merchant = deducted, like MDR). Each payment stores the charge it was made with. Listings run by Ultimate App have no charge and post to `platform_sales`.

# 5 URide

Ride-hailing, rentals with a driver, and self-drive rentals, with a driver wallet, live dispatch and full ledger posting. Library: `includes/uride.php` and `includes/uride_selfdrive.php`.

## 5.1 Vehicle types Boat added

Defined in `uride_vehicles()`. The flag `ride` allows point-to-point booking; `rent` allows hourly rental (with driver or self-drive). `uride_rentable_vehicles()` returns only rentable types.

| Key | Label | Seats | Ride | Rent | Self-drive license |
|---|---|---|---|---|---|
| `e_bike` | E-Bike | 1 rider | — | Yes (self-drive by default) | Verified ID only (Admin can require A/A1) |
| `motorcycle` | Motorcycle | 1 passenger | Yes | Yes | Code A / A1 |
| `e_trike` | E-Trike | 3 passengers | Yes | Yes | Code A / A1 |
| `car` | Car | 4 passengers | Yes | Yes | Code B |
| `van` | Van | 10–12 passengers | Yes | Yes | Code B / B1 |
| `boat` | **Boat** | Up to 10 | Yes | **No — point to point only** | — (never self-drive) |

> **Boat rules**
> Boat appears only in Book a Ride. Rental and self-drive quotes are always disabled, `uride_book_rental()` refuses it ("Boat is for point-to-point rides only"), fleet units ignore it, and Admin's rental and self-drive tables skip it. Only drivers registered with vehicle type Boat receive Boat requests. Operators enter their MARINA / boat captain license and the boat registration number.

## 5.2 Fares (Book a Ride)

Upfront and fixed at booking: **fare = max(minimum, base + per-km × (road km − km in base) + night surcharge)**, rounded up to the peso. Road km = straight-line distance × road factor (default 1.3). Set in **Admin › Settings › URide › Fare matrix**. Code defaults (placeholders):

| Vehicle | Base (1 km incl.) | Per km after | Minimum |
|---|---|---|---|
| Motorcycle | ₱40 | ₱12 | ₱40 |
| E-Trike | ₱30 | ₱15 | ₱30 |
| Car | ₱100 | ₱25 | ₱100 |
| Van | ₱300 | ₱40 | ₱300 |
| Boat | ₱500 | ₱100 | ₱500 |

## 5.3 Ride lifecycle and dispatch

**Requested** fare held (Credits/BCash) in `uride_escrow` → **Broadcast** online drivers of that type within the radius (3 km), GPS ≤ 2 min old → **Accepted** first accept wins (row lock) → **Arrived → In progress** live position for the passenger → **Completed** driver wallet + fare − commission

- Unaccepted requests expire after the timeout (default 5 min; rentals 10 min) and are refunded.
- Drivers need the minimum wallet (default ₱50) to go online; cash rides deduct the commission from the wallet.
- Commission: percentage (default 10%) or a fixed amount per trip, never more than the fare.
- Passengers rate the driver 1–5 stars. Drivers can skip a request or give an accepted ride back before pick-up.

## 5.4 Rent a Ride (with driver)

Packages of 1–12 hours or a whole day (default 8 h), now or scheduled up to 7 days ahead. Price = per hour × hours (at least the minimum), never more than the whole-day price. Scheduled rentals appear to drivers 60 minutes before the start.

| Vehicle | Per hour | Min hours | Whole day |
|---|---|---|---|
| E-Bike (off by default) | ₱150 | 1 | ₱800 |
| Motorcycle | ₱300 | 1 | ₱1,800 |
| E-Trike | ₱500 | 1 | ₱3,000 |
| Car | ₱800 | 2 | ₱5,000 |
| Van | ₱1,500 | 3 | ₱9,000 |

## 5.5 Self-drive and fleets

- The customer needs a verified ID (switchable), a driver's license on file that covers the vehicle and does not expire before the end, and must accept the self-drive rules. Paid with Credits or BCash; price plus refundable deposit are held.
- Owners enter how many units of each type they have ( `uride_fleet` ). Customers see "N available"; "Now" bookings are refused when nothing is free.
- Hand-over: the owner confirms the license and ID check; the rental clock starts. Return: good condition (deposit back), late (hourly rate per started hour after the grace period, default 15 min), or damage claim decided by Admin.

| Vehicle | Per hour | Min | Whole day | Deposit |
|---|---|---|---|---|
| E-Bike | ₱100 | 1 h | ₱600 | ₱1,000 |
| Motorcycle | ₱200 | 2 h | ₱1,200 | ₱3,000 |
| E-Trike | ₱350 | 2 h | ₱2,000 | ₱2,000 |
| Car | ₱600 | 4 h | ₱3,500 | ₱5,000 |
| Van | ₱1,200 | 4 h | ₱7,000 | ₱10,000 |

## 5.6 Places and maps

Pick-up and drop-off suggestions combine the customer's recent places, Admin-managed landmarks (**Admin › URide › URide places**), approved merchants near Boracay and Google Places. GPS "precise location" keeps the best fix for up to 20 s; pins are draggable and stored with 7 decimals.

## 5.7 Driver wallet

| Event | Driver wallet | Ledger |
|---|---|---|
| Ride completed (Credits/BCash) | + fare − commission | uride_escrow → driver_wallet + fee_revenue:uride |
| Ride completed (cash) | − commission | driver_wallet → fee_revenue:uride |
| Passenger cancellation fee | + fee | uride_escrow → user wallet (refund) + driver_wallet (fee) |
| Top-up: QR Ph / MCTC / BCash | + amount | paymongo_clearing / mctc_merchant_due / user_cash → driver_wallet |
| Payout requested → paid | − amount | driver_wallet → settlement_payable → bank_clearing |

# 6 Marketplace: UPass, UGo, ULocal, UEat

Merchants sell bookings and food through the Merchant Portal; Ultimate App can also run its own listings. Library: `includes/marketplace.php`, page helpers in `includes/market_page.php` and `includes/market_forms.php`.

## 6.1 Bookings (UPass, UGo, ULocal)

- Customer chooses date, start time (time slots with a guest capacity), guests or quantity, and for ULocal the address.
- Statuses: **pending → confirmed → completed**, or **declined / cancelled**. Listings can confirm instantly.
- Paid in the app (Credits or BCash) or on the day (when Pay Cash is on). Merchant bookings use `p2m_pay_tx()`; Ultimate App listings post to `platform_sales:<service>`.
- Each listing sets a free-cancellation cut-off (`cancel_hours` before the start).

## 6.2 UEat orders




- Restaurant profile: cuisine, cover, prep time, delivery fee, minimum order, hours (Philippine time), delivery and/or pick-up, open/close switch.
- Delivery is done by the restaurant itself; there is no rider dispatch yet.
- Merchant order board refreshes automatically; every status change creates a customer notification.

## 6.3 Moderation

**Admin › UGo · ULocal · UEat**: review new and edited listings and restaurants, manage any listing, feature listings, and confirm, complete or cancel bookings and orders (always with a full refund). Listing photos are stored in `storage/market/` and served by `market-photo.php`.

## 6.4 Agent commissions

Customers referred by an agent earn the agent a commission on completed URide trips, paid UEat orders and completed UPass/UGo/ULocal bookings (chapter 11.4).

# 7 Cancellation fees and reasons New

Customers must give a reason every time they cancel. A configurable percentage of what they paid in the app is kept as a fee when they cancel late. Library: `includes/cancellation.php`.

## 7.1 Settings

**Admin › Settings › Charges & limits › Cancellation fees** (0–100%, up to 2 decimals; 0% = free)

| Setting key | Default | Applies when | Fee goes to |
|---|---|---|---|
| `cancel.uride_pct` | 20% | After the free minutes once a driver accepted, or after the driver arrived | Driver wallet |
| `cancel.uride_free_min` | 2 min | Free window after the driver accepts | — |
| `cancel.booking_pct` | 25% | After the listing's cut-off and before the start | Merchant (or Ultimate App sales) |
| `cancel.ueat_pct` | 30% | While the order is accepted or preparing | Restaurant |

## 7.2 When customers can cancel

| Service | Free | With fee | Not allowed |
|---|---|---|---|
| URide | Searching; first free minutes after accept | After the free minutes; driver arrived | Trip started |
| UPass / UGo / ULocal | Pending; confirmed before the cut-off | Between the cut-off and the start | After the start |
| UEat | Before the restaurant accepts | Accepted or preparing | Ready or on the way |

- Only in-app payments (Credits or BCash) carry a fee. Cash rides and pay-on-the-day bookings never do.
- Declines by a driver, merchant or restaurant, and cancellations by Admin, are always refunded in full.
- Reasons come from `cancel_reasons($area)`; picking **Other** requires typed text. Reason and fee are stored (`cancel_reason`, `cancel_fee_centavos`) and shown to the customer, the driver or merchant (Orders › History) and Admin.

## 7.3 Ledger

- **URide:** `uride_release_hold($pdo,$ride,'cancelled',$fee)` moves held − fee back to the customer's wallet and the fee to `driver_wallet:ID` (group `RF-…`); the driver's wallet history shows "Cancellation fee".
- **Merchant bookings and UEat:** full refund first, then `mk_cancel_fee_to_merchant_tx()` moves the fee from the customer's wallet to `merchant_payable:ID` (group `CF-…`).
- **Ultimate App listings:** only paid − fee is refunded; the fee stays in `platform_sales`.

# 8 Wallet, transfers and exchange

## 8.1 Buy Credits

- **QR Ph (PayMongo):** in-app QR via Payment Intent ( `topup-qr.php` ), with hosted checkout as fallback. Credited only by a signed webhook ( `payment.paid` or `checkout_session.payment.paid` ) or by an authenticated status check that matches session, reference, mode and exact amount. QR expires after 30 minutes. A ₱5.00 service fee per top-up is configurable.
- **MCTC (cash):** the customer shows a request QR at an MCTC top-up center (an approved merchant). The merchant scans it, enters a receipt number (unique per merchant) and confirms; Credits are added immediately and the cash becomes the merchant's MCTC due, deducted from the next payout.

## 8.2 BCash

- Convert Credits ⇄ BCash 1:1, no fee, atomic with a unique request key.
- **Send / Receive / My QR:** BCash QR codes (with the BCash logo) always send BCash; Credits QR codes always pay Credits. ₱1–₱10,000 per transfer, max 20 sends per hour, no fee.
- When KYC is required for customers, unverified accounts cannot send Credits or BCash or use UCash In.

## 8.3 P2M and P2P payments

Scan QR detects merchant QR codes ( `UM-…` , P2M) and person QR codes ( `UA-…` , P2P). P2M shows the charge and total before paying, with Credits or BCash.

## 8.4 Currency exchange

- USD/PHP is the main pair; about 20 more currencies can be enabled in **Admin › ERP › Currency exchange**. Each currency has its own wallet and decimals.
- Rates: Frankfurter first, then currency-api, or a manual rate. Refreshed hourly ( `cron/fx-refresh.php` ). Exchanges pause if the rate is older than 24 h; an automatic rate outside ₱20–200 or more than 10% from the last one is refused.
- Fees 0–5% for buy and sell (default 0.50% for USD, 1.00% for other currencies). The amount is locked to the rate shown.

# 9 Bills, load and cash-in (SUNIWAY)

| Tile | Purpose | SUNIWAY group |
|---|---|---|
| **UBills** | Electricity (AKELCO), water, internet, cable, SSS, Pag-IBIG, PhilHealth… | `bills` |
| **ULoad** | Prepaid load for Globe, TM, Smart, TNT, DITO | `eload` |
| **UCash In** | Send Credits to GCash, Maya, ShopeePay and other e-wallets | `ecCash` |

**GET providers**
billers, networks, e-wallets
→
**POST transactions/preview**
provider fee and total shown
→
**Debit Credits**
then `POST transactions`
→
**GET transactions/{id}**
until final

- **Successful:** done. **Rejected / Failed:** automatic refund. **Processing:** rechecked; refunded if it later fails.
- **Checking** (no reply in time): Credits stay on hold; Finance resolves it in **Admin › Bills & Load › Needs checking** (Refund or Mark successful).
- Ledger: debit `user_credits`, credit `suniway_payable` (amount + provider fee) and `fee_revenue:<service>` (optional convenience fee).
- The Partner API key is encrypted in settings. "Test connection" counts the billers, products and e-wallets found.

# 10 Identity: KYC, liveness and Face Login

## 10.1 KYC

- Available to customers, merchants, drivers and agents: ID front and back, then a guided liveness selfie (look straight, left, right, up, down, blink), then review and send.
- Face detection runs on the phone with MediaPipe. The side to turn to flashes red; correct poses turn green and capture automatically; wrong turns are counted and shown to Admin.
- The "look straight" selfie is compared with the ID photo using face-api. The server re-checks the descriptor distance (default max 0.60) and refuses a mismatch.
- Every submission is reviewed by a person in **Admin › KYC verification**: Approve, Ask to resubmit, or Reject. "Manual" submissions (no automatic check) need extra care.
- Rules (Admin › KYC › Rules): merchants, drivers and agents need approved KYC before approval (on by default); customers only for sending money and UCash In (off by default).

## 10.2 Face Login (customers)

- Set up in Profile: a short liveness check whose face must match the approved KYC face. The phone gets an HttpOnly, SameSite=Strict device key; only its SHA-256 is stored.
- Login: the server issues random steps and a one-time code valid for 2 minutes, checks step order and timing, and compares the 128-number face descriptor (default max distance 0.45).
- 5 wrong tries lock Face Login on that phone until a password login. Users can turn it off per phone or on all phones.
- Face Login is a convenience login for the person's own phone, not a bank-grade identity proof. The password always works.

> **Privacy**
> Only still photos (KYC) or 128 numbers (Face Login) leave the phone. Photos are re-encoded (no EXIF/GPS), stored with random names in `storage/kyc/`, and shown only to signed-in admins with KYC access. IP address and browser are recorded for audit.

# 11 Portals

## 11.1 Admin Panel and roles

| Role | Access |
|---|---|
| **Super Admin** | Everything, including settings, rates, driver approval and admin users |
| **Finance** | Payments and refunds, finance, ledger, settlements, reports, Credits adjustments, driver and agent payouts, currency exchange |
| **Support** | Customers, merchant and driver review, tickets, notes, KYC decisions, ride cancellation |
| **Barangay Officer** | UBarangay applications for their own barangay only |

Every admin action is recorded in the audit log. First-time setup ( `/admin/` ) creates the Super Admin and then disables itself.

## 11.2 Merchant Portal and MCTC

- Onboarding: business, owner login, registration numbers, documents, payout account, map pin, KYC → Admin review → Approved.
- After approval: payment QR (optional fixed amount), payments and CSV, payouts, listings (UPass/UGo/ULocal/UEat), orders and bookings, support tickets.
- MCTC Top-up Center (on for every approved merchant by default): scan a customer's or driver's top-up QR, collect cash, enter the receipt and confirm. Cash collected is owed to Ultimate App ( `merchants.mctc_due_centavos` , ledger `mctc_merchant_due:ID` ) and deducted from the next payout, or recorded as a remittance.

## 11.3 URide Driver App

- Registration with selfie, license, vehicle (including Boat), plate or boat registration, documents; status Submitted → Under review / Needs info → Approved.
- Online/offline toggle, live request feed with beep, accept/skip, navigation, trip steps, self-drive hand-over and return, fleet units, wallet with QR Ph / MCTC / BCash top-ups and payouts.
- The app must stay open with location allowed to receive requests (browsers pause background GPS).

## 11.4 Agents Portal

- Approved agents get a referral link ( `/register.php?ref=AG…` ), code and QR. Customers are tagged once at sign-up, for `agents.earn_days` (default 365).
- Default rates: URide 2% of the fare, UEat 3% (earned after a 3-day hold), UPass/UGo/ULocal 5% (earned when completed). Paid by Ultimate App as an expense.
- Master Agents and Sub-Agents (one level). Masters earn an override on top (default 0.5% URide/UEat, 1% bookings).

## 11.5 UBarangay

Permits paid with Credits, BCash or QR Ph, reviewed by the Barangay Officer, issued with a verification QR ( `/verify.php` ). Fees post to `brgy_payable` . Off by default.

# 12 Security

| Control | Implementation |
|---|---|
| Passwords | `password_hash()` / `password_verify()`; separate logins for customers, drivers, merchants, agents and admins |
| CSRF | Token per session, checked by `verify_csrf()` on every POST |
| Rate limits | `rate_limit_enforce(action, actor, limit, window)` backed by the `rate_limits` table (logins, bookings, cancellations, transfers…) |
| SQL injection | PDO prepared statements; dynamic identifiers come from fixed allow-lists |
| XSS | All output escaped with `e()`; user text such as cancellation reasons is always escaped |
| Clickjacking | `X-Frame-Options: DENY`; strict Content-Security-Policy on file viewers and camera pages |
| Private files | `storage/` and `config/` blocked by `.htaccess`; files served only through permission-checked scripts with sandbox CSP |
| Secrets | API keys encrypted in `app_settings`; never logged or shown after saving; never commit `config/config.php`, `config/app_secret.php` or database dumps |
| Webhooks | Signature and timestamp (5 min) verification; idempotent settlement; no raw bodies logged |
| Audit | Admin audit log; ledger entries are append-only |

> **Full package zips contain secrets**
> The full deployment zip includes the live `config/config.php` (database password). Share it only with the server owner, and never upload it to a public place.

# 13 Deployment

## 13.1 Server requirements

- PHP 8.1+ with PDO MySQL, cURL, GD (image re-encoding), OpenSSL, mbstring; outbound HTTPS.
- MySQL 8 or MariaDB 10.x with InnoDB (tested on MariaDB 10.11).
- HTTPS (needed for camera, GPS, PWA install and Apple sign-in).
- Writable `storage/` (or the folder in `config/storage.php`) and, once, `config/`.

## 13.2 Steps (Hostinger)

1. Back up the database and the site files (including `config/` and `storage/`).
2. Import migrations in phpMyAdmin (13.3). Every migration is safe to run again.
3. Upload the update zip (changed files only) or the full zip. Keep the live `config/config.php`, `config/app_secret.php` and `storage/`.
4. Configure in Admin › Settings: PayMongo keys, Google Maps key, URide fares, payment methods, cancellation fees, SUNIWAY.
5. Set the PayMongo webhook to `https://YOUR-DOMAIN/webhook.php` (events `payment.paid` and `checkout_session.payment.paid`).
6. Add the cron job: `php /home/<user>/public_html/cron/fx-refresh.php` every hour.
7. Smoke-test on real phones: booking, payment, driver app, merchant orders.

## 13.3 Migration order

| # | File | Adds |
|---|---|---|
| 1 | `schema.sql` | Base tables (new install only) |
| 2 | `paymongo, mctc, fees, conversion, service_requests, notifications, ueat_orders, boracay_cash, boracay_cash_reverse, oauth, rate_limits` | Early platform features (see each module guide) |
| 3 | `ALL_UPDATES_2026-09.sql` | News, QR payments, Admin platform, UBarangay, in-app QR Ph, URide driver app, MCTC merchants |
| 4 | `agents → agents_team` | Agents Portal, Master/Sub-Agents |
| 5 | `suniway` | UBills, ULoad, UCash In |
| 6 | `kyc` | Identity verification |
| 7 | `bcash_transfers` | BCash Send/Receive |
| 8 | `fx → fx_currencies` | Currency exchange |
| 9 | `uride_places → uride_rental → uride_selfdrive` | Places, Rent a Ride, Self-drive |
| 10 | `face` | Face match and Face Login |
| 11 | `marketplace` | UGo, ULocal, UEat merchants |
| 12 | `uride_fleet → upass_credits` | E-Bike, Van, fleets; UPass bookings |
| 13 | `pay_methods` | Pay Cash switch, BCash payments |
| 14 | `cancellation` | Cancellation fees and reasons |
| 15 | `uride_boat` | Boat vehicle type and fares |

## 13.4 Configuration files

| File / variable | Purpose |
|---|---|
| `config/config.php · DB_HOST, DB_NAME, DB_USER, DB_PASS` | Database connection (environment variables override) |
| `config/paymongo.php · PAYMONGO_*` | Mode, keys, webhook secrets, `app_url` |
| `config/app_secret.php` | Generated encryption key for stored secrets — back it up |
| `config/storage.php` | Optional private uploads folder outside `public_html` |
| `APP_URL, GOOGLE_*, FACEBOOK_*, APPLE_*` | Social sign-in |

# 14 Testing

CLI test suites run real database transactions against a **copy** of the database and delete everything they create. Never run them on production or against real provider APIs.

```
DB_HOST=localhost DB_NAME=copy_db DB_USER=... DB_PASS=... php tests/uride_boat_test.php
```

| Suite | Covers |
|---|---|
| `uride_test.php` | Fares, night surcharge, commission, top-ups (QR Ph, MCTC, BCash), dispatch radius, first-accept-wins, refunds, payouts, reconciliation |
| `uride_rental_test.php` · `uride_selfdrive_test.php` | Rentals, scheduling, licenses, deposits, late fees, damage claims |
| `uride_fleet_test.php` · `uride_places_test.php` | Fleets and availability; place search |
| `uride_boat_test.php` | Boat ride-only rules, fares, dispatch to boat operators, payment |
| `cancellation_test.php` | Fee rules, free windows, reasons, ledger for URide, bookings, UEat |
| `pay_methods_test.php` | Pay Cash switch, BCash payments and refunds |
| `marketplace_test.php` · `p2m_test.php` · `qr_pay_test.php` | Bookings, UEat orders, merchant payments, P2P |
| `boracay_cash_test.php` · `bcash_transfer_test.php` | Conversions and transfers |
| `fx_test.php` | Exchange math, rates, fees, limits (needs `fx_fake_api.php` on port 8097) |
| `suniway_test.php` | Bills, load, cash-in (needs `suniway_fake_api.php` on port 8098) |
| `paymongo_test.php` | Webhook signatures and settlement ( `PM_FAKE_API=1` ) |
| `kyc_test.php` · `face_test.php` | KYC rules and Face Login |
| `agents_test.php` · `agents_team_test.php` | Commissions, teams, payouts |
| `mctc_test.php` · `mctc_merchant_test.php` | Cash top-up approvals |
| `ubarangay_test.php` | Permits, payments, certificates |

```
# fake provider servers for local tests
php -S 127.0.0.1:8098 tests/suniway_fake_api.php &
php -S 127.0.0.1:8097 tests/fx_fake_api.php &
```

> **Status at this edition**
> All 23 suites passed on 11–13 October 2026 (PHP 8.3, MariaDB), together with browser checks of the admin settings, URide booking (including Boat), and the cancel forms.

# 15 Operations and known limits

## 15.1 Routine operations

- **Daily:** process driver, agent and merchant payouts (mark paid with the transfer reference); resolve SUNIWAY "Needs checking"; review KYC and driver applications.
- **Weekly:** check Admin › Finance reconciliation (wallets, escrow, MCTC dues); export reports.
- **Backups:** database, `config/` (especially `app_secret.php`) and the storage folder, together.
- **Rates and prices:** starter fares, rental prices, deposits and listing prices are placeholders — confirm them with the LGU, TODA and partners.

## 15.2 Known limits

- URide distances use straight line × road factor, not routed distance (this also applies to Boat routes).
- UEat delivery is done by the restaurant; no rider dispatch yet.
- Extending a running rental is not built in.
- Liveness and Face Login run on the phone; every KYC is still reviewed by a person.
- QR Ph and bank/e-wallet payouts are manual (Finance marks them paid); no automated payout provider is connected.
- BCT, Buy BCT and Markets pages are "coming soon".
- Stored-value and e-money rules (BSP Circular 1166) should be reviewed with counsel before BCash is offered as redeemable cash.

## 15.3 Related guides in the package

URIDE-SETUP.md · URIDE-RENTAL-SETUP.md · URIDE-SELFDRIVE-SETUP.md · URIDE-PLACES-SETUP.md · URIDE-BOAT-SETUP.md · CREDITS-AND-RENTAL-FLEET-SETUP.md · MARKETPLACE-SETUP.md · PAYMENT-METHODS-SETUP.md · CANCELLATION-FEES-SETUP.md · BORACAY-CASH-SETUP.md · BCASH-SECTION-SETUP.md · USD-EXCHANGE-SETUP.md · SUNIWAY-SETUP.md · KYC-SETUP.md · FACE-LIVENESS-LOGIN-SETUP.md · AGENTS-PORTAL-SETUP.md · ADMIN-MERCHANT-SETUP.md · MERCHANT-PORTAL-SETUP.md · MCTC-SETUP.md · PAYMONGO-SETUP.md · UBARANGAY-SETUP.md · PWA-SETUP.md · MENU-COMMUNITY-SETUP.md · README.md